

ControlCheck AI

Control Rule Catalogue v0.2

MVP deterministic control rules | 20 rules | 5 control domains

Purpose. This catalogue defines the first production-oriented rule set for ControlCheck AI. The rule engine must calculate deterministic findings from canonical project data; LLM reasoning explains and prioritizes findings but does not replace rule calculations.

1. Rule Set Summary

Data Quality	5	DQ-001 to DQ-005	Ensure the data can be trusted before project-control analysis.
Cost	5	CST-001 to CST-005	Detect current and emerging cost exposure.
Schedule	5	SCH-001 to SCH-005	Detect slippage, critical-path risk, and float issues.
Progress	2	PRG-001 to PRG-002	Detect progress inconsistencies and performance gaps.
Cross-domain	3	CST-006, PRG-003, XDOM-001	Detect interactions between spend, progress, and schedule.

2. Evaluation Principles

- Rules are deterministic and testable. The same input data and analysis date must produce the same finding set.
- Every material finding must carry evidence references to source rows and canonical records whenever technically possible.
- Thresholds are configuration, not hard-coded business logic. Each rule should expose project-level overrides.
- Findings must be idempotent within the same project, rule version, and reporting cut-off; repeated runs must not create uncontrolled duplicates.
- LLM output is advisory. It can summarize evidence, propose root causes, and recommend actions, but it cannot invent metrics or silently alter rule results.

3. Severity Model

Critical	Immediate management review. Material cost/schedule exposure or severe data integrity issue.	Red
Warning	Requires project-control review and likely action, but impact is not yet confirmed material.	Amber
Observation	Useful signal or data hygiene issue. No direct escalation required unless repeated.	Blue/Neutral

4. Detailed Rule Catalogue

01. DQ-001 — Missing WBS Reference

Category	data_quality
Default severity	warning
Objective	Detect budget, cost, schedule, or progress records that cannot be tied to a project WBS node.
Inputs	budget_records, cost_records, schedule_activities, progress_records, wbs_nodes
Detection logic	For each domain record, wbs_node_id must resolve to an active WBS node for the same project. Missing or unresolved WBS creates a finding.
Threshold	Any unresolved WBS record; configurable minimum count = 1.
Evidence	Source dataset, original row number, source WBS value, project code.
Finding template	{dataset} contains {count} records without a valid WBS reference.
Impact	Weak traceability; cost/progress/schedule rollups may be incomplete or misleading.
Recommendation	Map the source WBS to the canonical WBS tree or correct the source record before analysis.
False-positive handling	Allow approved non-WBS records to be excluded by dataset mapping rule.
Acceptance criteria	A test file with 5 unresolved WBS records produces exactly 5 evidence rows and 1 grouped finding.

02. DQ-002 — Duplicate Cost Transaction

Category	data_quality
Default severity	warning
Objective	Detect probable duplicate cost transactions introduced by repeated rows or repeated file imports.
Inputs	cost_records, raw_rows
Detection logic	Build a normalized duplicate key from project_id + transaction_date + vendor_code/name + po_number + document_number + amount + normalized description. Group count > 1.
Threshold	Duplicate group count >= 2; confidence increases when document_number or PO matches exactly.
Evidence	Duplicate record IDs, source row numbers, duplicate key fields, amounts.
Finding template	{count} probable duplicate cost transactions detected across {groups} duplicate groups.
Impact	Can inflate actual cost, commitments, vendor exposure, and forecasts.
Recommendation	Verify source duplicates against invoice/document number and remove or mark repeated rows.
False-positive handling	Permit legitimate split postings where document number differs or accounting status indicates separate posting.
Acceptance criteria	Known duplicate pair is detected as one group; legitimate split pair is not grouped when document numbers differ.

03. DQ-003 — Contradictory Date or Progress Values

Category	data_quality
Default severity	warning
Objective	Identify records with internally inconsistent dates or progress values before control calculations.
Inputs	raw_rows, schedule_activities, progress_records
Detection logic	Flag schedule rows where finish < start, actual finish < actual start, progress outside 0..1, or progress period is earlier than project planned start when disallowed.
Threshold	Any contradictory record; ingestion should reject hard-invalid values and retain evidence.
Evidence	Dataset, row number, offending fields, validation message.
Finding template	{count} records contain contradictory date/progress values.
Impact	Can distort delay, SPI, progress, and health calculations.
Recommendation	Correct source values and re-import the affected dataset.
False-positive handling	Project-specific exceptions must be explicitly configured rather than silently ignored.
Acceptance criteria	Each invalid record is traceable to the original row and receives a deterministic validation reason.

04. DQ-004 — Orphan WBS Reference

Category	data_quality
Default severity	critical
Objective	Detect records referencing a WBS code that does not exist in the project canonical WBS tree.
Inputs	raw_rows, wbs_nodes and domain datasets
Detection logic	Source WBS value is present but no matching canonical wbs_nodes row exists for the same project.
Threshold	Any orphan WBS; severity critical when orphaned amount exceeds 1% of project budget.
Evidence	Source WBS, project, affected records, aggregated amount/progress.
Finding template	WBS {source_wbs} is not mapped to the project WBS tree; {count} records are affected.
Impact	Rollup leakage; affected costs/progress may disappear from management totals.
Recommendation	Create a mapping or add the missing WBS node after project-control approval.
False-positive handling	None for active analysis; retired WBS can be supported through explicit alias/mapping.
Acceptance criteria	A missing WBS with material cost is critical and appears in the evidence view.

05. DQ-005 — Vendor Identity Inconsistency

Field	Value
Category	data_quality
Default severity	observation
Objective	Detect the same vendor appearing under multiple codes/names that can fragment spend analysis.
Inputs	cost_records
Detection logic	Normalize vendor_name. Flag a normalized name mapped to >1 vendor_code, or one vendor_code mapped to materially different normalized names.
Threshold	At least 2 distinct identities; warning when affected spend > 2% of project actual cost.
Evidence	Vendor codes, names, transaction count, total amount.
Finding template	Vendor identity inconsistency found for {normalized_vendor}; {identity_count} identities represent {amount}.
Impact	Vendor concentration and cost-driver analysis can be understated or duplicated.
Recommendation	Create a canonical vendor mapping and standardize future imports.
False-positive handling	Joint ventures or legal entities may legitimately have separate codes; permit approved mapping exceptions.
Acceptance criteria	Known alias set is grouped together while explicitly distinct legal entities remain separate when configured.

06. CST-001 — Actual Cost Exceeds Budget

Field	Value
Category	cost
Default severity	critical
Objective	Detect WBS-level actual cost overruns against approved budget.
Inputs	cost_records where cost_type=actual, budget_records
Detection logic	Aggregate actual amount and approved budget by project + WBS. Variance = actual - approved_budget. Trigger when variance > 0.
Threshold	Default: any positive variance; severity critical when variance_pct >= 10% or variance_amount >= configurable materiality.
Evidence	Budget record(s), all contributing actual transactions, WBS rollup.
Finding template	WBS {wbs} actual cost exceeds approved budget by {variance_amount} ({variance_pct}).
Impact	Confirmed cost overrun at current reporting cut-off.
Recommendation	Review quantity/price variance, commitments, change orders, and coding before escalation.
False-positive handling	Approved revised budget must replace the prior baseline used for the check.
Acceptance criteria	Synthetic WBS with budget 100 and actual 124 returns variance 24 and 24%.

07. CST-002 — Actual + Commitment Exceeds Budget

Field	Value
Category	cost
Default severity	critical
Objective	Detect likely future overrun where actual plus open commitment already exceeds approved budget.
Inputs	cost_records where cost_type in (actual, commitment), budget_records
Detection logic	Exposure = actual + commitment. Trigger when exposure > approved_budget.
Threshold	Warning when > 100%; critical when >= 110% or materiality threshold is exceeded.
Evidence	Budget, actual transactions, commitment transactions, PO numbers.
Finding template	WBS {wbs} has committed exposure of {exposure} versus budget {budget}.
Impact	Forward-looking overrun exposure even if actual spend has not yet exceeded budget.
Recommendation	Review open PO values, remaining scope, pending variations, and potential decommitment.
False-positive handling	Exclude commitments already cancelled/closed or fully reversed.
Acceptance criteria	Open commitment pushes exposure from 98% to 106%; rule returns warning with correct evidence.

08. CST-003 — Cost Spike vs Recent Run Rate

Field	Value
Category	cost
Default severity	warning
Objective	Detect sudden cost acceleration that warrants project-control review.
Inputs	cost_records actual by month/period
Detection logic	For each WBS, current_period_cost > trailing_3_period_average * (1 + spike_threshold). Only evaluate when at least 3 prior comparable periods exist.
Threshold	Default spike_threshold = 30%; configurable minimum absolute amount = 1% of BAC or project materiality.
Evidence	Current period cost, three-period baseline, ratio, underlying transactions.
Finding template	WBS {wbs} cost increased {increase_pct} versus trailing 3-period average.
Impact	Potential acceleration, bulk procurement, coding issue, or emerging cost risk.
Recommendation	Inspect high-value transactions, PO releases, procurement timing, and coding changes.
False-positive handling	Planned bulk purchases can be tagged as expected events.
Acceptance criteria	A 40% spike above three-period average is detected; 20% spike is not.

09. CST-004 — Vendor Concentration Risk

Field	Value
Category	cost
Default severity	warning
Objective	Detect excessive dependence on one vendor within project spend.
Inputs	cost_records actual by vendor
Detection logic	$\text{vendor_share} = \text{vendor_actual} / \text{project_actual}$ for selected period or cumulative view.
Threshold	Default warning $\geq 40\%$; critical $\geq 60\%$ when project configuration enables concentration monitoring.
Evidence	Vendor total, project total, share, top transactions.
Finding template	Vendor {vendor} represents {share} of project actual cost.
Impact	Commercial concentration and change/claim dependency may be high.
Recommendation	Review scope dependency, outstanding commitments, commercial exposure, and alternatives.
False-positive handling	Single-source contracts can be explicitly marked expected.
Acceptance criteria	Vendor with 45% share creates warning; 35% does not.

10. CST-005 — High-Value Transaction Outlier

Field	Value
Category	cost
Default severity	observation
Objective	Surface unusually large individual transactions relative to WBS or project budget.
Inputs	cost_records actual, budget_records
Detection logic	$\text{transaction_share} = \text{transaction_amount} / \text{WBS_budget}$. Trigger when share exceeds threshold.
Threshold	Default $\geq 5\%$ of WBS budget or $\geq 2\%$ of project budget, with configurable absolute materiality floor.
Evidence	Transaction record, budget context, PO/document number, vendor.
Finding template	Transaction {document} represents {share} of WBS budget and requires review.
Impact	Large transactions may contain quantity, price, coding, or timing anomalies.
Recommendation	Verify supporting document, approved scope, coding, and period allocation.
False-positive handling	Known milestone or advance-payment transactions can be tagged expected.
Acceptance criteria	Transaction at 6% of WBS budget is surfaced; 3% is not unless project materiality overrides.

11. CST-006 — High Cost with Low Progress

Field	Value
Category	cross_domain
Default severity	critical
Objective	Detect WBSs where spend is materially advanced while physical progress remains materially low.
Inputs	cost_records actual, budget_records, progress_records
Detection logic	For same WBS and latest period: cost_pct = actual_to_date / budget; progress_pct = actual_progress. Trigger when cost_pct >= 80% AND progress_pct <= 50%.
Threshold	Default 80% cost vs <=50% progress; configurable by project type.
Evidence	Budget, actual-to-date, latest progress, period history, major transactions.
Finding template	WBS {wbs} has consumed {cost_pct} of budget while progress is {progress_pct}.
Impact	Potential quantity/price issue, low productivity, front-loaded procurement, or scope accounting risk.
Recommendation	Perform cost-to-progress reconciliation and validate quantity, productivity, and procurement timing.
False-positive handling	Front-loaded procurement packages should be tagged as expected.
Acceptance criteria	80% cost / 45% progress triggers; 75% / 45% does not.

12. SCH-001 — Overdue Activity

Field	Value
Category	schedule
Default severity	warning
Objective	Detect incomplete activities whose baseline finish is before the analysis date.
Inputs	schedule_activities
Detection logic	actual_progress < 1 AND baseline_finish < analysis_date AND status not in completed/cancelled.
Threshold	Default any overdue activity; critical when activity is critical_flag=true or overdue_days >= 14.
Evidence	Activity, baseline finish, actual progress/status, analysis date, overdue days.
Finding template	Activity {activity} is overdue by {days} days with progress {progress}.
Impact	Schedule slippage and potential downstream impact.
Recommendation	Confirm actual field status, recovery plan, and downstream dependencies.
False-positive handling	Approved baseline changes must update the active baseline or be versioned explicitly.
Acceptance criteria	Activity 10 days overdue creates warning; 14 days or critical activity creates critical.

13. SCH-002 — Baseline Finish Slippage

Field	Value
Category	schedule
Default severity	warning
Objective	Detect completed activities that finished later than baseline.
Inputs	schedule_activities
Detection logic	actual_finish > baseline_finish. slippage_days = actual_finish - baseline_finish.
Threshold	Default warning >= 7 days; critical >= 14 days or project-configured threshold.
Evidence	Activity baseline and actual dates, slippage days.
Finding template	Activity {activity} finished {slippage_days} days later than baseline.
Impact	Evidence of schedule variance and possible milestone impact.
Recommendation	Capture delay reason, responsibility, and lesson/action if material.
False-positive handling	Approved rebaseline should be compared using the correct baseline version.
Acceptance criteria	8-day slippage is warning; 15-day slippage is critical.

14. SCH-003 — Critical Activity Delay

Field	Value
Category	schedule
Default severity	critical
Objective	Prioritize delayed activities on the critical path.
Inputs	schedule_activities
Detection logic	critical_flag=true AND (overdue OR actual_finish > baseline_finish OR total_float_days < 0).
Threshold	Any critical activity meeting delay condition.
Evidence	Activity critical flag, dates, float, progress, status.
Finding template	Critical activity {activity} is delayed/at risk and may affect project completion.
Impact	Potential direct impact to project completion or major milestone.
Recommendation	Escalate recovery plan, predecessors/successors, resources, and mitigation actions.
False-positive handling	Critical flag must reflect the approved current schedule version.
Acceptance criteria	A critical activity 1 day overdue creates a critical finding regardless of amount.

15. SCH-004 — Activity Progress Lag

Field	Value
Category	schedule
Default severity	warning
Objective	Detect activities whose actual progress materially trails planned progress.
Inputs	schedule_activities
Detection logic	progress_lag = planned_progress - actual_progress. Trigger when lag exceeds threshold.
Threshold	Default >= 10 percentage points; critical >= 20 points for critical activities.
Evidence	Activity planned vs actual progress, dates, status.
Finding template	Activity {activity} is {lag_pp} percentage points behind plan.
Impact	Early signal of schedule recovery requirement.
Recommendation	Validate field progress and determine root cause and recovery action.
False-positive handling	Progress methodology changes must be reflected before analysis.
Acceptance criteria	11 pp lag triggers warning; 9 pp does not.

16. SCH-005 — Negative Total Float

Field	Value
Category	schedule
Default severity	critical
Objective	Detect schedule activities with negative total float.
Inputs	schedule_activities
Detection logic	total_float_days < 0.
Threshold	Any negative float.
Evidence	Activity, float value, schedule version/date.
Finding template	Activity {activity} has {float_days} days negative float.
Impact	Activity requires schedule recovery or network review to protect milestones.
Recommendation	Review driving path, constraints, calendars, and recovery options.
False-positive handling	External interface constraints may be intentionally modeled; retain evidence and approval notes.
Acceptance criteria	Float -1 creates critical finding; float 0 does not.

17. PRG-001 — WBS Progress Lag

Field	Value
Category	progress
Default severity	warning
Objective	Detect WBS-level actual progress materially behind plan for the latest reporting period.
Inputs	progress_records
Detection logic	$\text{progress_lag} = \text{planned_progress} - \text{actual_progress}$ for latest period per WBS.
Threshold	Default ≥ 10 percentage points; critical ≥ 20 points or when linked to critical schedule activities.
Evidence	Latest progress record, prior periods, WBS.
Finding template	WBS {wbs} is {lag_pp} percentage points behind plan.
Impact	Project delivery risk if gap persists.
Recommendation	Investigate workforce constraints, productivity, materials, resources, and schedule logic.
False-positive handling	Milestone-weighted or earned-value methodologies should be configured consistently.
Acceptance criteria	12 pp gap creates warning; 8 pp gap does not.

18. PRG-002 — Progress Above 100 Percent

Field	Value
Category	progress
Default severity	warning
Objective	Detect source progress values that exceed logical maximums.
Inputs	raw_rows, progress_records, schedule_activities
Detection logic	Flag any source actual_progress or planned_progress > 1.0 , including values encoded as 100+ percent after normalization.
Threshold	Any value $> 100\%$.
Evidence	Original source value, normalized value, row number.
Finding template	Progress value {value} for {wbs/activity} exceeds 100%.
Impact	Can distort project progress and health calculations.
Recommendation	Correct source encoding and re-import.
False-positive handling	None in canonical percentage field; alternate units must be transformed during mapping.
Acceptance criteria	101% is flagged; 100% is valid.

19. PRG-003 — Cost Rising While Progress Flat

Category	cross_domain
Default severity	critical
Objective	Detect WBS periods where cost continues to rise but physical progress barely changes.
Inputs	cost_records actual by period, progress_records
Detection logic	For latest two comparable periods: cost_change_pct >= 20% AND progress_change_pp <= 2 points.
Threshold	Default cost +20% and progress change <=2 pp; materiality floor applies.
Evidence	Two-period cost totals, progress values, top transactions.
Finding template	WBS {wbs} cost increased {cost_change_pct} while progress changed only {progress_change_pp}.
Impact	Possible productivity loss, stalled workforce, advance procurement, or coding issue.
Recommendation	Reconcile cost to installed quantities/progress and investigate top spend drivers.
False-positive handling	Advance procurement and known front-loaded packages can be tagged expected.
Acceptance criteria	Cost +25% with progress +1 pp triggers; cost +15% does not.

20. XDOM-001 — Schedule Delay with High Cost Exposure

Category	cross_domain
Default severity	critical
Objective	Prioritize WBSs where schedule delay and cost exposure coexist.
Inputs	schedule_activities, budget_records, cost_records, progress_records
Detection logic	Trigger when WBS has overdue/critical delayed activity AND cost_exposure_pct >= 80% of budget OR actual_cost > 90% of budget.
Threshold	Default schedule delay + >=80% exposure; critical when exposure >=100%.
Evidence	Delayed activities, budget, actual/commitment exposure, progress.
Finding template	WBS {wbs} combines schedule delay with {exposure_pct} cost exposure.
Impact	Higher probability of time and cost overrun occurring together.
Recommendation	Escalate WBS as integrated cost-schedule risk; review recovery, remaining budget, and forecast.
False-positive handling	Planned front-loaded work must be marked expected and still monitored.
Acceptance criteria	Delayed activity + 85% exposure creates critical finding; delayed + 60% exposure does not.

5. Control Rule Runtime Contract

Every rule is stored in control_rules.expression as JSONB. The backend should compile this declarative expression into a safe evaluator rather than executing arbitrary Python supplied by users.

```

{
  "metric": "actual_to_date",
  "group_by": [
    "project_id",
    "wbs_node_id"
  ],
  "conditions": [
    {
      "field": "actual_to_date",
      "op": ">",
      "value": "approved_budget"
    }
  ],
  "threshold": {
    "type": "percent",
    "value": 10
  },
  "severity_map": {
    "warning": 0,
    "critical": 10
  },
  "evidence": [
    "budget_records",
    "cost_records"
  ]
}

```

6. Threshold Configuration

spike_threshold_pct	30	CST-003	Cost acceleration threshold vs trailing average.
vendor_concentration_pct	40	CST-004	Vendor share warning threshold.
high_value_wbs_pct	5	CST-005	Single transaction share of WBS budget.
low_progress_pct	50	CST-006	Upper progress threshold for high-cost/low-progress.
high_cost_pct	80	CST-006	Cost consumed threshold for high-cost/low-progress.
overdue_days	14	SCH-001	Critical overdue threshold.
slippage_warning_days	7	SCH-002	Schedule finish slippage warning threshold.
slippage_critical_days	14	SCH-002	Schedule finish slippage critical threshold.
activity_progress_lag_pp	10	SCH-004	Activity planned-vs-actual progress warning gap.
wbs_progress_lag_pp	10	PRG-001	WBS planned-vs-actual progress warning gap.
cost_rise_pct	20	PRG-003	Cost increase threshold for flat-progress detection.
progress_change_pp	2	PRG-003	Maximum progress change considered flat.
exposure_pct	80	XDOM-001	Cost exposure threshold when schedule delay exists.

7. Evidence Requirements

- Every finding stores evidence JSON and should preferably reference evidence_records for normalized links.

- At minimum, evidence must identify project, rule, WBS/activity where applicable, source dataset, source file, and original row number.
- Aggregated metrics must store enough information to reproduce the calculation: numerator, denominator, period, and threshold used.
- For cost findings, include the contributing transaction IDs or raw row IDs when the set is not excessively large. Large evidence sets may be summarized with a downloadable detail view.

8. Test Data Strategy

The next engineering artifact should be a synthetic project dataset deliberately seeded with known violations. Each rule needs at least one positive case and one near-boundary negative case.

DQ-001	5 records with missing WBS	5 evidence rows
CST-001	Budget 100 / actual 124	24 overrun and 24%
CST-003	Baseline average 100 / current 140	40% spike
SCH-003	Critical activity overdue 1 day	Critical finding
PRG-003	Cost +25% / progress +1 pp	Cross-domain finding
XDOM-001	Delayed + 85% cost exposure	Critical integrated risk

9. Implementation Order

1. Implement DQ-001..DQ-005 first so downstream calculations operate on trusted data.
2. Implement CST-001..CST-005 and SCH-001..SCH-005 next; these provide the first valuable dashboard signals.
3. Add PRG-001..PRG-002, then cross-domain rules CST-006, PRG-003, XDOM-001.
4. Only after deterministic findings pass the synthetic test suite should the AI reasoning layer be connected.
5. Every production release must record rule version changes and preserve historical finding calculations.

10. Definition of Done for Rule Engine v0.1

- 20 rules implemented behind a common rule runtime contract.
- Rule thresholds configurable by project without changing code.
- All rules return deterministic findings with severity, metric, and evidence.
- Synthetic dataset produces expected findings for positive and negative test cases.
- Rule execution records timing and counts in control_rule_runs.
- Finding results are auditable back to source file and row where applicable.
- Re-running the same input is idempotent for the same project, cut-off, rule version, and dataset snapshot.
- No LLM is required for numeric calculation or threshold evaluation.

Runtime Alignment Supplement v0.2

This supplement is generated from data/controlcheck_rule_catalogue_v0.2.json, the governed runtime catalogue consumed by the deterministic engine. It records the evaluation grain and active thresholds for all 20 MVP rules.

DQ-001	Missing WBS Reference	transaction	is_null	minimum_count=1
DQ-002	Duplicate Cost Transaction	transaction	group_count_gte	duplicate_count_min=2
DQ-003	Contradictory Date or Progress Values	activity	any_invalid	progress_min=0; progress_max=1
DQ-004	Orphan WBS Reference	transaction	not_in_master	minimum_count=1
DQ-005	Vendor Identity Inconsistency	transaction	normalized_name_mismatch	variant_count_min=2

CST-001	Actual Cost Exceeds Budget	wbs	gt	actual_budget_ratio_min=1.0
CST-002	Actual + Commitment Exceeds Budget	wbs	gt	exposure_budget_ratio_min=1.0
CST-003	Cost Spike vs Recent Run Rate	wbs	all_gte	run_rate_multiplier_min=1.3; project_budget_share_min=0.01
CST-004	Vendor Concentration Risk	vendor_wbs	gte	warning_share_min=0.4; critical_share_min=0.6
CST-005	High-Value Transaction Outlier	transaction	all_gte	wbs_share_min=0.25; project_share_min=0.03
CST-006	High Cost with Low Progress	wbs	all	cost_pct_min=0.8; progress_pct_max=0.5
SCH-001	Overdue Activity	activity	overdue	overdue_days_min=1; critical_days_min=14
SCH-002	Baseline Finish Slippage	activity	gte	slippage_days_min=7; critical_days_min=14
SCH-003	Critical Activity Delay	activity	critical_and_delayed	delay_days_min=1
SCH-004	Activity Progress Lag	activity	gte	lag_pp_min=0.1; critical_lag_pp_min=0.2
SCH-005	Negative Total Float	activity	lt	float_days_max_exclusive=0
PRG-001	WBS Progress Lag	wbs	gte	lag_pp_min=0.1; critical_lag_pp_min=0.2
PRG-002	Progress Above 100 Percent	wbs	gt	progress_max=1.0
PRG-003	Cost Rising While Progress Flat	period_wbs	all	cost_change_pct_min=0.2; progress_change_pp_max=0.02; current_period_project_budget_min=0.01
XDOM-001	Schedule Delay with High Cost Exposure	wbs	schedule_and_cost	exposure_pct_min=0.8; actual_pct_alterate_min=0.9

Governed clarifications

- CST-004 Vendor Concentration Risk is evaluated at vendor + WBS grain in v0.2, preventing unrelated WBS spend from masking local concentration.
- CST-005 High-Value Transaction Outlier requires both inclusive materiality gates: 25% of WBS budget and 3% of project budget.
- PRG-003 Cost Rising While Progress Flat requires current-period spend materiality of at least 1% of project budget in addition to its cost/progress movement conditions.
- All calculations remain deterministic. LLM output may explain and prioritize findings but cannot change thresholds, metrics, or evidence.
- Every persisted result records the catalogue version and SHA-256 so an analysis run can be reproduced against the same governed definition.